A Q8.24 Framework for Online Neural Equalization in Millimeter-Wave Analog Radio-over-Fiber Links

Author information to be added

Working draft: provisional abstract and partial introduction. Quantitative equalizer results and FPGA/power claims are not yet included.

Abstract

Analog radio-over-fiber links at millimeter-wave frequencies require equalization methods that address component nonlinearities while supporting practical receiver implementation. This work investigates a framework for online neural equalization of a single-carrier 16-QAM optical-beating link near 60 GHz. A modular physical model represents electrical drive, Mach–Zehnder modulation, fiber propagation, photodetection, wireless transmission, and receiver recovery. Independent training, validation, and test sequences support comparisons between a least-mean-square feed-forward equalizer and neural receivers with configurable hidden-layer dimensions and activation functions. The neural formulation uses continuous in-phase and quadrature outputs, mean-square-error training, and chronological sample-by-sample stochastic gradient descent. All stored neural quantities use signed Q8.24 arithmetic, with explicit rounding and saturation boundaries and backward propagation based on pre-update weights. The evaluation protocol links bit-error rate, error-vector magnitude, and convergence behavior to the resource, throughput, and energy requirements of a candidate FPGA implementation. Initial training sequences and periodic supervised bursts provide the intended mechanism for adaptation, with the first hardware phase operating at a controlled sample rate. The framework is designed to identify operating conditions in which nonlinear neural compensation provides a useful advantage and to assess the implementation cost of maintaining that advantage during online training. **Index terms**—Analog radio-over-fiber, millimeter-wave communications, neural equalization, online training, fixed-point arithmetic, FPGA.

1 Introduction

Analog radio-over-fiber (A-RoF) provides a means of transporting radio signals between centralized processing and remote radio units while retaining an analog optical interface. At millimeter-wave frequencies, its optical and wireless segments must be considered together because the recovered waveform depends on the cascaded component responses. Modulation and amplification can introduce nonlinear distortion, while filtering, fiber propagation, and receiver noise also affect signal quality. Consequently, equalizer performance should be assessed using a physical model that distinguishes linear channel effects from nonlinear component behavior and identifies the assumptions underlying its noise and power budget [1].

A feed-forward equalizer (FFE) trained using the least-mean-square (LMS) algorithm offers a useful adaptive linear baseline. Its finite impulse response can compensate linear distortion within the available observation window, and its coefficient updates can be implemented using regular multiply–accumulate operations. However, a linear mapping cannot in general reproduce the inverse of a nonlinear component characteristic. This limitation motivates the investigation of neural equalizers when amplitude compression or modulation nonlinearities produce signal-dependent distortion. Any resulting benefit must be established against a suitably tuned baseline under the same physical conditions and with independent evaluation data.

Previous work has demonstrated the relevance of neural nonlinear compensation in millimeter-wave RoF systems. Liu *et al.* investigated a complex-valued multilevel ANN equalizer for 16-QAM and 64-QAM transmission in a 60-GHz RoF link, addressing cross-modulation between the in-phase and quadrature components [2]. In a separate 60-GHz experiment using BPSK, Liu *et al.* studied an adaptive activation function and compared its convergence and equalization performance with conventional nonlinear compensation [3]. These studies motivate examining both the nonlinear receiver mapping and its activation function. Their experimental configurations differ, so their numerical results cannot be transferred directly to the link considered here.

Building on this motivation, the present study considers neural equalization together with the arithmetic and control required for online training. In a hardware implementation, the activation function determines both the forward evaluation and the derivative used during weight adaptation. Network dimensions, numerical precision, rounding, saturation, and update ordering therefore affect the implemented learning process as well as the inference datapath. The research objective is to identify useful BER or EVM improvements over LMS equalization and then assess whether the selected neural configurations can preserve those improvements within a consistent Q8.24 training and inference formulation. The initial implementation target is a sequential FPGA engine operating at a controlled sample rate; processing the full communication waveform at its symbol rate is a separate architectural objective. *To be completed: veri-*

fied contribution statement, broader hardware-related literature review, and paper organization.

2 Physical Link and Impairment Model

Planned: component models, parameters, calibration provenance, impairment assumptions, and numerical consistency.

3 Adaptive Equalizers and Training Protocol

Planned: LMS baseline, neural configurations, supervised framing, independent datasets, and validation/test protocol.

4 Q8.24 Training and Inference Architecture

Planned: arithmetic, activation/derivative, memory, phase control, and Python/RTL agreement.

5 Communication Performance

Pending: independent-test BER/EVM, convergence, activation comparison, error counts, and limitations.

6 FPGA Implementation and Live Adaptation

Pending: KV260 implementation, controlled-rate adaptation, resources, latency, throughput, and overload handling.

7 Power and Energy Evaluation

Pending: tool estimates, measurement methodology, measured power, accepted-operation counts, and energy.

8 Discussion and Conclusion

To be written from the completed evidence.

References

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